Taller - Sesion 5 - Series de Tiempo y Python IIE - UNAM

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Abstract Este Notebook incluye una introduccion al manejo de Series de Tiempo con Python

```
import numpy as np
import pandas as pd
from pandas import read_excel # funcion para leer archivos de excel
from io import StringIO
import contextlib
import re
import matplotlib.pyplot as plt

# statsmodels
import statsmodels
from statsmodels.tsa.statespace.sarimax import SARIMAX
from statsmodels.tsa.stattools import adfuller
from statsmodels.tsa.stattools import kpss
from statsmodels.graphics.tsaplots import plot_acf, plot_pacf
from statsmodels.tsa.seasonal import seasonal_decompose


import pmdarima as pm
from pmdarima import ARIMA
from pmdarima import auto_arima

import sktime
from sktime import datasets
from sktime.utils import plot_series

import warnings
warnings.filterwarnings("ignore")

import session_info
```

```
#!pip install xlrd
#!pip install sktime
#!pip install seaborn
```

```
h=30
```

```
df = pd.read_csv('AirPassengers.csv')
df1 = read_excel('ClayBricks.xls')
df2 = read_excel('Electricity.xls')
df3 = read_excel('MilkProduction.xls')
df4 = read_excel('JapaneseCars.xls')
df5 = read_excel('HouseSales.xls')
```

Serie 1

```
dates=pd.to_datetime(df["Month"])
ts=pd.Series(np.array(df["#Passengers"]),index=dates)
```

```
ts
```

```
Month
1949-01-01    112
1949-02-01    118
1949-03-01    132
1949-04-01    129
1949-05-01    121
             ...
1960-08-01    606
1960-09-01    508
1960-10-01    461
1960-11-01    390
1960-12-01    432
Length: 144, dtype: int64
```

```
from pmdarima.datasets import load_wineind

# this is a dataset from R
wineind = load_wineind().astype(np.float64)
```

```
len(wineind)
```

```
176
```

```
stepwise_fit = pm.auto_arima(wineind, start_p=1, start_q=1,
                             max_p=3, max_q=3, m=12, start_P=0,
                             seasonal=True,
                             d=1, D=1, trace=True,
                             error_action='ignore',  # don't want to know if
an order does not work
                             suppress_warnings=True,  # don't want
convergence warnings
                             stepwise=True)  # set to stepwise
```

```
Performing stepwise search to minimize aic
 ARIMA(1,1,1)(0,1,1)[12]             : AIC=3066.492, Time=0.26 sec
 ARIMA(0,1,0)(0,1,0)[12]             : AIC=3131.408, Time=0.02 sec
 ARIMA(1,1,0)(1,1,0)[12]             : AIC=3097.884, Time=0.07 sec
 ARIMA(0,1,1)(0,1,1)[12]             : AIC=3066.329, Time=0.08 sec
 ARIMA(0,1,1)(0,1,0)[12]             : AIC=3089.456, Time=0.03 sec
 ARIMA(0,1,1)(1,1,1)[12]             : AIC=3067.457, Time=0.14 sec
 ARIMA(0,1,1)(0,1,2)[12]             : AIC=3067.481, Time=0.22 sec
 ARIMA(0,1,1)(1,1,0)[12]             : AIC=3071.631, Time=0.08 sec
 ARIMA(0,1,1)(1,1,2)[12]             : AIC=inf, Time=1.44 sec
 ARIMA(0,1,0)(0,1,1)[12]             : AIC=3117.921, Time=0.14 sec
 ARIMA(0,1,2)(0,1,1)[12]             : AIC=3065.533, Time=0.10 sec
 ARIMA(0,1,2)(0,1,0)[12]             : AIC=3087.883, Time=0.04 sec
 ARIMA(0,1,2)(1,1,1)[12]             : AIC=3066.239, Time=0.19 sec
 ARIMA(0,1,2)(0,1,2)[12]             : AIC=3066.373, Time=0.35 sec
 ARIMA(0,1,2)(1,1,0)[12]             : AIC=3070.728, Time=0.10 sec
 ARIMA(0,1,2)(1,1,2)[12]             : AIC=inf, Time=1.07 sec
 ARIMA(1,1,2)(0,1,1)[12]             : AIC=3066.424, Time=0.25 sec
 ARIMA(0,1,3)(0,1,1)[12]             : AIC=3066.351, Time=0.28 sec
 ARIMA(1,1,3)(0,1,1)[12]             : AIC=3068.295, Time=0.50 sec
 ARIMA(0,1,2)(0,1,1)[12] intercept   : AIC=3066.787, Time=0.12 sec

Best model:  ARIMA(0,1,2)(0,1,1)[12]
Total fit time: 5.521 seconds
```

```
df_fit = pm.auto_arima(ts, start_p=1, start_q=1,
                             max_p=3, max_q=3, m=12, start_P=0,
                             seasonal=True,
                             d=1, D=1, trace=True,
                             error_action='ignore',  # don't want to know if
an order does not work
                             suppress_warnings=True,  # don't want
```

```
convergence warnings
                              stepwise=True)  # set to stepwise
```

```
Performing stepwise search to minimize aic
 ARIMA(1,1,1)(0,1,1)[12]             : AIC=1022.896, Time=0.15 sec
 ARIMA(0,1,0)(0,1,0)[12]             : AIC=1031.508, Time=0.02 sec
 ARIMA(1,1,0)(1,1,0)[12]             : AIC=1020.393, Time=0.06 sec
 ARIMA(0,1,1)(0,1,1)[12]             : AIC=1021.003, Time=0.08 sec
 ARIMA(1,1,0)(0,1,0)[12]             : AIC=1020.393, Time=0.03 sec
 ARIMA(1,1,0)(2,1,0)[12]             : AIC=1019.239, Time=0.17 sec
 ARIMA(1,1,0)(2,1,1)[12]             : AIC=inf, Time=1.22 sec
 ARIMA(1,1,0)(1,1,1)[12]             : AIC=1020.493, Time=0.19 sec
 ARIMA(0,1,0)(2,1,0)[12]             : AIC=1032.120, Time=0.21 sec
 ARIMA(2,1,0)(2,1,0)[12]             : AIC=1021.120, Time=0.34 sec
 ARIMA(1,1,1)(2,1,0)[12]             : AIC=1021.032, Time=0.33 sec
 ARIMA(0,1,1)(2,1,0)[12]             : AIC=1019.178, Time=0.20 sec
 ARIMA(0,1,1)(1,1,0)[12]             : AIC=1020.425, Time=0.07 sec
 ARIMA(0,1,1)(2,1,1)[12]             : AIC=inf, Time=1.10 sec
 ARIMA(0,1,1)(1,1,1)[12]             : AIC=1020.327, Time=0.25 sec
 ARIMA(0,1,2)(2,1,0)[12]             : AIC=1021.148, Time=0.23 sec
 ARIMA(1,1,2)(2,1,0)[12]             : AIC=1022.805, Time=0.40 sec
 ARIMA(0,1,1)(2,1,0)[12] intercept   : AIC=1021.017, Time=0.44 sec

Best model:  ARIMA(0,1,1)(2,1,0)[12]
Total fit time: 5.497 seconds
```

```
df_fit.summary()
```

Dep. Variable:	y	No. Observations:	144
Model:	SARIMAX(0, 1, 1)x(2, 1, [], 12)	Log Likelihood	−505.589
Date:	Sat, 08 Nov 2025	AIC	1019.178
Time:	05:24:00	BIC	1030.679
Sample:	01-01-1949	HQIC	1023.851
	• 12-01-1960		
Covariance Type:	opg		

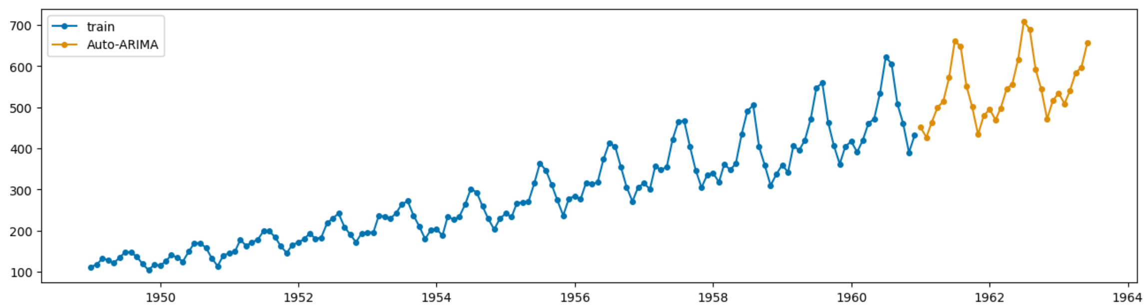
	coef	std err	z	P>|z|	[0.025	0.975]
ma.L1	−0.3634	0.074	−4.945	0.000	−0.508	−0.219
ar.S.L12	−0.1239	0.090	−1.372	0.170	−0.301	0.053
ar.S.L24	0.1911	0.107	1.783	0.075	−0.019	0.401

sigma2	130.4480	15.527	8.402	0.000	100.016	160.880

Ljung-Box (L1) (Q):	0.01	Jarque-Bera (JB):	4.59
Prob(Q):	0.92	Prob(JB):	0.10
Heteroskedasticity (H):	2.70	Skew:	0.15
Prob(H) (two-sided):	0.00	Kurtosis:	3.87

```
ts_pre=df_fit.predict(h)
```

```
plot_series(ts,ts_pre,labels=["train","Auto-ARIMA"])
```



Serie 2

```
dates1=pd.to_datetime(df1["Dates"])
ts1=pd.Series(np.array(df1["Bricks"]),index=dates1)
```

```
ts1
```

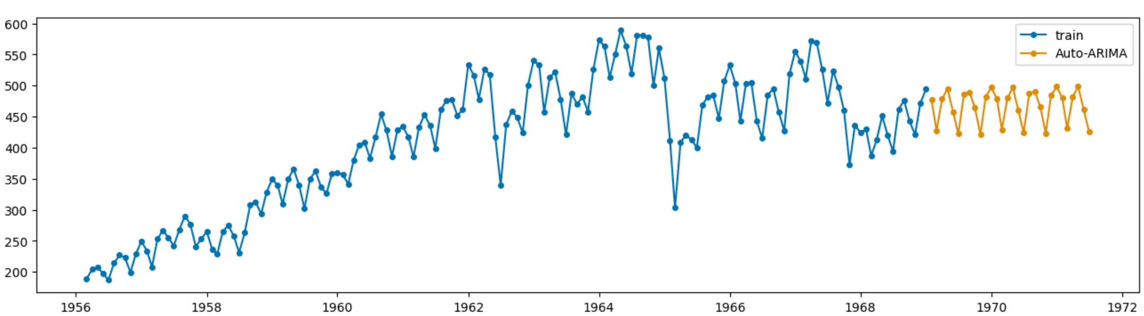
```
Dates
1956-03-01    189
1956-04-01    204
1956-05-01    208
1956-06-01    197
1956-07-01    187
             ...
1968-09-01    476
1968-10-01    443
1968-11-01    421
1968-12-01    472
1969-01-01    494
Length: 155, dtype: int64
```

```
df1_fit = pm.auto_arima(ts1, start_p=1, start_q=1,
                              max_p=4, max_q=4, m=12, start_P=0,
                              seasonal=True,
                              d=1, D=1, trace=True,
                              error_action='ignore',  # don't want to know if
an order does not work
                              suppress_warnings=True,  # don't want
convergence warnings
                              stepwise=True)  # set to stepwise
```

```
Performing stepwise search to minimize aic
 ARIMA(1,1,1)(0,1,1)[12]             : AIC=1323.673, Time=0.32 sec
 ARIMA(0,1,0)(0,1,0)[12]             : AIC=1366.878, Time=0.02 sec
 ARIMA(1,1,0)(1,1,0)[12]             : AIC=1336.158, Time=0.08 sec
 ARIMA(0,1,1)(0,1,1)[12]             : AIC=1320.197, Time=0.15 sec
 ARIMA(0,1,1)(0,1,0)[12]             : AIC=1365.182, Time=0.04 sec
 ARIMA(0,1,1)(1,1,1)[12]             : AIC=1321.846, Time=0.30 sec
 ARIMA(0,1,1)(0,1,2)[12]             : AIC=1321.857, Time=0.45 sec
 ARIMA(0,1,1)(1,1,0)[12]             : AIC=1337.557, Time=0.08 sec
 ARIMA(0,1,1)(1,1,2)[12]             : AIC=inf, Time=1.69 sec
 ARIMA(0,1,0)(0,1,1)[12]             : AIC=1320.760, Time=0.08 sec
 ARIMA(0,1,2)(0,1,1)[12]             : AIC=1321.569, Time=0.20 sec
 ARIMA(1,1,0)(0,1,1)[12]             : AIC=1319.848, Time=0.12 sec
 ARIMA(1,1,0)(0,1,0)[12]             : AIC=1364.159, Time=0.02 sec
 ARIMA(1,1,0)(1,1,1)[12]             : AIC=1321.518, Time=0.20 sec
 ARIMA(1,1,0)(0,1,2)[12]             : AIC=1321.535, Time=0.43 sec
 ARIMA(1,1,0)(1,1,2)[12]             : AIC=inf, Time=1.39 sec
 ARIMA(2,1,0)(0,1,1)[12]             : AIC=1321.381, Time=0.17 sec
 ARIMA(2,1,1)(0,1,1)[12]             : AIC=1323.363, Time=0.31 sec
 ARIMA(1,1,0)(0,1,1)[12] intercept   : AIC=1321.729, Time=0.16 sec

Best model:  ARIMA(1,1,0)(0,1,1)[12]
Total fit time: 6.199 seconds
```

```
ts1_pre=df1_fit.predict(h)
plot_series(ts1,ts1_pre,labels=["train","Auto-ARIMA"])
```



Serie 3

```
dates2=pd.to_datetime(df2["Month and year"])
ts2=pd.Series(np.array(df2["Kwh"]),index=dates2)
```

```
ts2
```

```
Month and year
1956-01-01     1254
1956-02-01     1290
1956-03-01     1379
1956-04-01     1346
1956-05-01     1535
              ...
1995-04-01    13032
1995-05-01    14268
1995-06-01    14473
1995-07-01    15359
1995-08-01    14457
Length: 476, dtype: int64
```

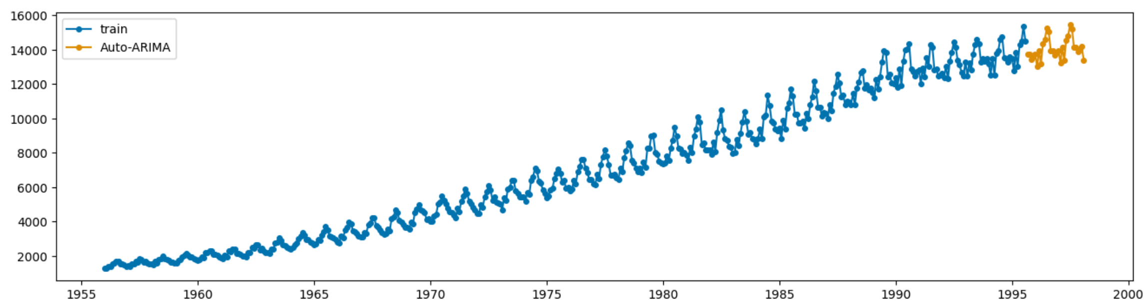
```
df2_fit = pm.auto_arima(ts2, start_p=1, start_q=1,
                              max_p=4, max_q=4, m=12, start_P=0,
                              seasonal=True,
                              d=1, D=1, trace=True,
                              error_action='ignore',  # don't want to know if
an order does not work
                              suppress_warnings=True,  # don't want
convergence warnings
                              stepwise=True)  # set to stepwise
```

```
Performing stepwise search to minimize aic
 ARIMA(1,1,1)(0,1,1)[12]             : AIC=6065.732, Time=0.56 sec
 ARIMA(0,1,0)(0,1,0)[12]             : AIC=6401.709, Time=0.03 sec
 ARIMA(1,1,0)(1,1,0)[12]             : AIC=6188.054, Time=0.19 sec
 ARIMA(0,1,1)(0,1,1)[12]             : AIC=6064.304, Time=0.41 sec
 ARIMA(0,1,1)(0,1,0)[12]             : AIC=6221.772, Time=0.08 sec
 ARIMA(0,1,1)(1,1,1)[12]             : AIC=6064.317, Time=0.55 sec
 ARIMA(0,1,1)(0,1,2)[12]             : AIC=6063.199, Time=1.87 sec
 ARIMA(0,1,1)(1,1,2)[12]             : AIC=6050.038, Time=5.17 sec
 ARIMA(0,1,1)(2,1,2)[12]             : AIC=6051.909, Time=5.06 sec
 ARIMA(0,1,1)(2,1,1)[12]             : AIC=6058.186, Time=1.21 sec
 ARIMA(0,1,0)(1,1,2)[12]             : AIC=6226.586, Time=1.12 sec
 ARIMA(1,1,1)(1,1,2)[12]             : AIC=6058.202, Time=6.80 sec
 ARIMA(0,1,2)(1,1,2)[12]             : AIC=6055.825, Time=5.88 sec
```

```
 ARIMA(1,1,0)(1,1,2)[12]             : AIC=inf, Time=3.59 sec
 ARIMA(1,1,2)(1,1,2)[12]             : AIC=inf, Time=5.61 sec
 ARIMA(0,1,1)(1,1,2)[12] intercept   : AIC=inf, Time=4.30 sec

Best model:  ARIMA(0,1,1)(1,1,2)[12]
Total fit time: 42.456 seconds
```

```
ts2_pre=df2_fit.predict(h)
plot_series(ts2,ts2_pre,labels=["train","Auto-ARIMA"])
```



Serie 4

```
df3
```

	Monthly Milk Production per Cow
0	589
1	561
2	640
3	656
4	727
…	…
163	858
164	817
165	827
166	797
167	843

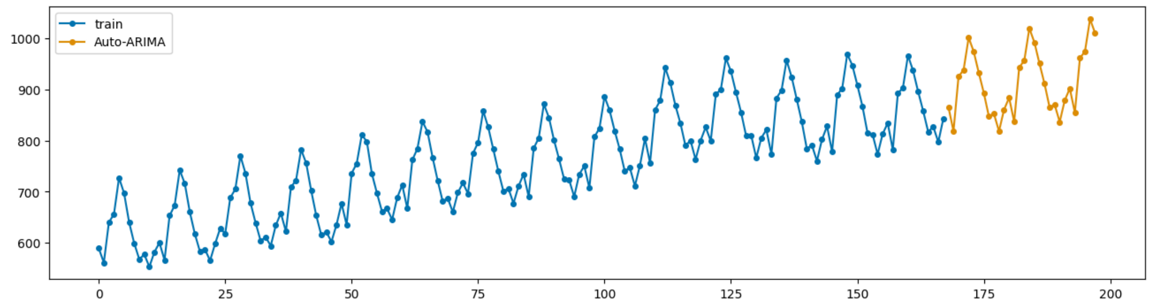
```python
#dates3=pd.to_datetime(df3["Monthly Milk Production per Cow"])
ts3=pd.Series(np.array(df3["Monthly Milk Production per Cow"]))
```

```python
df3_fit = pm.auto_arima(ts3, start_p=1, start_q=1,
                              max_p=4, max_q=4, m=12, start_P=0,
                              seasonal=True,
                              d=1, D=1, trace=True,
                              error_action='ignore',  # don't want to know if
an order does not work
                              suppress_warnings=True,  # don't want
convergence warnings
                              stepwise=True)  # set to stepwise
```

```
Performing stepwise search to minimize aic
 ARIMA(1,1,1)(0,1,1)[12]             : AIC=1068.064, Time=0.14 sec
 ARIMA(0,1,0)(0,1,0)[12]             : AIC=1119.969, Time=0.01 sec
 ARIMA(1,1,0)(1,1,0)[12]             : AIC=1081.584, Time=0.05 sec
 ARIMA(0,1,1)(0,1,1)[12]             : AIC=1066.296, Time=0.10 sec
 ARIMA(0,1,1)(0,1,0)[12]             : AIC=1114.995, Time=0.02 sec
 ARIMA(0,1,1)(1,1,1)[12]             : AIC=1068.030, Time=0.15 sec
 ARIMA(0,1,1)(0,1,2)[12]             : AIC=1067.976, Time=0.39 sec
 ARIMA(0,1,1)(1,1,0)[12]             : AIC=1082.123, Time=0.07 sec
 ARIMA(0,1,1)(1,1,2)[12]             : AIC=inf, Time=1.21 sec
 ARIMA(0,1,0)(0,1,1)[12]             : AIC=1072.280, Time=0.07 sec
 ARIMA(0,1,2)(0,1,1)[12]             : AIC=1067.796, Time=0.13 sec
 ARIMA(1,1,0)(0,1,1)[12]             : AIC=1066.207, Time=0.10 sec
 ARIMA(1,1,0)(0,1,0)[12]             : AIC=1114.845, Time=0.02 sec
 ARIMA(1,1,0)(1,1,1)[12]             : AIC=1067.913, Time=0.14 sec
 ARIMA(1,1,0)(0,1,2)[12]             : AIC=1067.857, Time=0.27 sec
 ARIMA(1,1,0)(1,1,2)[12]             : AIC=inf, Time=0.87 sec
 ARIMA(2,1,0)(0,1,1)[12]             : AIC=1067.922, Time=0.13 sec
 ARIMA(2,1,1)(0,1,1)[12]             : AIC=1066.696, Time=0.45 sec
 ARIMA(1,1,0)(0,1,1)[12] intercept   : AIC=1068.200, Time=0.13 sec

Best model:  ARIMA(1,1,0)(0,1,1)[12]
Total fit time: 4.484 seconds
```

```python
ts3_pre=df3_fit.predict(h)
plot_series(ts3,ts3_pre,labels=["train","Auto-ARIMA"])
```



Serie 5

```
df4
```

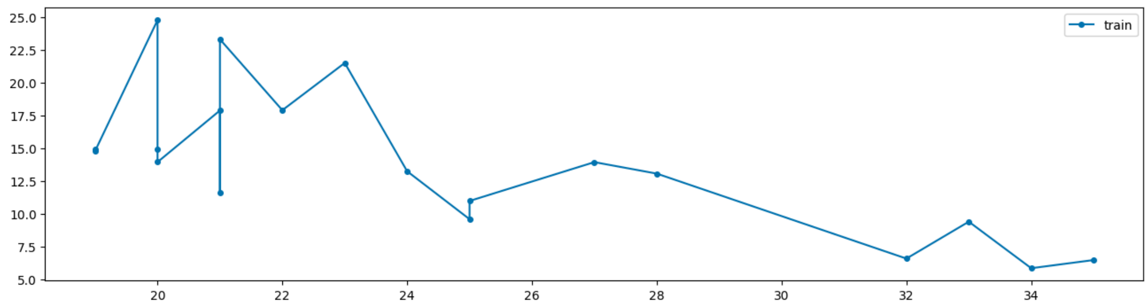
	Mileage	Price
0	19	14.944
1	19	14.799
2	20	24.760
3	20	14.929
4	20	13.949
5	21	17.879
6	21	11.650
7	21	23.300
8	22	17.899
9	23	21.498
10	24	13.249
11	25	9.599
12	25	10.989
13	27	13.945
14	28	13.071
15	32	6.599
16	33	9.410
17	34	5.866
18	35	6.488

```
dates4=pd.to_numeric(df4["Mileage"])
ts4=pd.Series(np.array(df4["Price"]),index=dates4)
```

```
ts4
```

```
Mileage
19    14.944
19    14.799
20    24.760
20    14.929
20    13.949
21    17.879
21    11.650
21    23.300
22    17.899
23    21.498
24    13.249
25     9.599
25    10.989
27    13.945
28    13.071
32     6.599
33     9.410
34     5.866
35     6.488
dtype: float64
```

```
plot_series(ts4,labels=["train"])
```



Serie 6

```
df5
```

	Month	HouseSales
0	1956-01-01	55
1	1956-02-01	60
2	1956-03-01	68
3	1956-04-01	63
4	1956-05-01	65
…	…	…
270	1978-07-01	64
271	1978-08-01	63
272	1978-09-01	55
273	1978-10-01	54
274	1978-11-01	44

```
dates5=pd.to_datetime(df5["Month"])
ts5=pd.Series(np.array(df5["HouseSales"]),index=dates5)
```

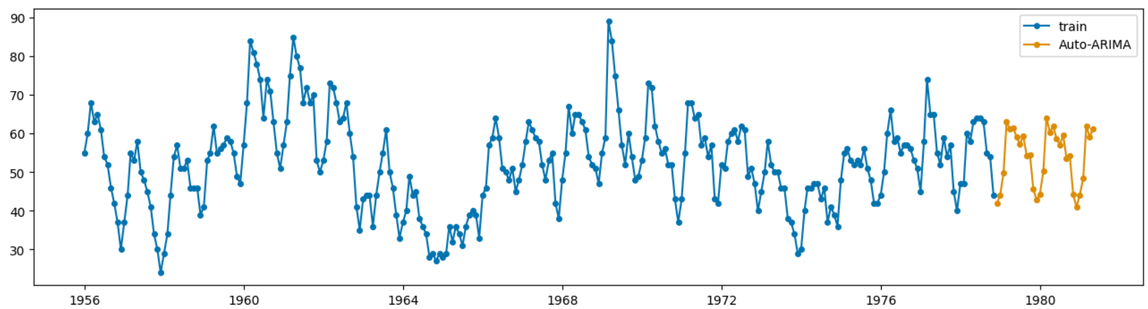
```
df5_fit = pm.auto_arima(ts5, start_p=1, start_q=1,
                              max_p=4, max_q=4, m=12, start_P=0,
                              seasonal=True,
                              d=1, D=1, trace=True,
                              error_action='ignore',  # don't want to know if
an order does not work
                              suppress_warnings=True,  # don't want
convergence warnings
                              stepwise=True)  # set to stepwise
```

```
Performing stepwise search to minimize aic
 ARIMA(1,1,1)(0,1,1)[12]             : AIC=inf, Time=0.73 sec
 ARIMA(0,1,0)(0,1,0)[12]             : AIC=1690.202, Time=0.02 sec
 ARIMA(1,1,0)(1,1,0)[12]             : AIC=1633.997, Time=0.07 sec
 ARIMA(0,1,1)(0,1,1)[12]             : AIC=inf, Time=0.40 sec
 ARIMA(1,1,0)(0,1,0)[12]             : AIC=1689.378, Time=0.03 sec
 ARIMA(1,1,0)(2,1,0)[12]             : AIC=1619.080, Time=0.26 sec
 ARIMA(1,1,0)(2,1,1)[12]             : AIC=inf, Time=2.49 sec
 ARIMA(1,1,0)(1,1,1)[12]             : AIC=inf, Time=0.73 sec
 ARIMA(0,1,0)(2,1,0)[12]             : AIC=1618.152, Time=0.32 sec
 ARIMA(0,1,0)(1,1,0)[12]             : AIC=1634.148, Time=0.07 sec
 ARIMA(0,1,0)(2,1,1)[12]             : AIC=inf, Time=1.25 sec
 ARIMA(0,1,0)(1,1,1)[12]             : AIC=inf, Time=0.35 sec
 ARIMA(0,1,1)(2,1,0)[12]             : AIC=1618.780, Time=0.30 sec
```

```
 ARIMA(1,1,1)(2,1,0)[12]             : AIC=1615.016, Time=0.60 sec
 ARIMA(1,1,1)(1,1,0)[12]             : AIC=1628.633, Time=0.17 sec
 ARIMA(1,1,1)(2,1,1)[12]             : AIC=inf, Time=2.22 sec
 ARIMA(1,1,1)(1,1,1)[12]             : AIC=inf, Time=0.77 sec
 ARIMA(2,1,1)(2,1,0)[12]             : AIC=1615.740, Time=0.87 sec
 ARIMA(1,1,2)(2,1,0)[12]             : AIC=1615.917, Time=0.64 sec
 ARIMA(0,1,2)(2,1,0)[12]             : AIC=1616.640, Time=0.37 sec
 ARIMA(2,1,0)(2,1,0)[12]             : AIC=1618.350, Time=0.37 sec
 ARIMA(2,1,2)(2,1,0)[12]             : AIC=1617.223, Time=1.46 sec
 ARIMA(1,1,1)(2,1,0)[12] intercept   : AIC=1616.932, Time=2.32 sec

Best model:  ARIMA(1,1,1)(2,1,0)[12]
Total fit time: 16.849 seconds
```

```
ts5_pre=df5_fit.predict(h)
plot_series(ts5,ts5_pre,labels=["train","Auto-ARIMA"])
```




```
session_info.show(html=False)
```

```
-----
matplotlib          3.9.4
numpy               2.0.2
pandas              2.3.1
pmdarima            2.0.4
session_info        v1.0.1
sktime              0.38.5
statsmodels         0.14.5
-----
IPython             8.18.1
jupyter_client      8.6.3
jupyter_core        5.8.1
-----
Python 3.9.23 | packaged by conda-forge | (main, Jun  4 2025, 17:57:12) [GCC
13.3.0]
Linux-6.8.0-87-generic-x86_64-with-glibc2.39
```

```
-----
Session information updated at 2025-11-08 05:26
```